

Kamal Pangeni, PhD

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I am a data scientist with 7+ years of industry experience in predictive analytics, machine learning, and GenAI. Trained as a theoretical physicist, I enjoy solving complex problems with innovative ideas. I am passionate about developing and delivering AI/ML solutions using cutting-edge technologies to enhance, augment, and automate company-wide decision-making processes.

EXPERIENCE

Senior Data Scientist, CVS Health

Oct 2021 - Present

- Developed an LLM Agent to assess members' eligibility for prior authorization approval based on electronic documents from healthcare providers. Preliminary results indicate that the agent successfully approved cases **with 95% precision, streamlining the prior authorization authentication process.**
- Developed a Generative AI-based application utilizing the LangChain framework to expedite the creation of question sets from policy documents, **reducing the turnaround time from several days to less than an hour.**
- Developed anomaly detection algorithm to identify pricing irregularities in generic drugs that certain pharmacies exploited. Collaborated closely with the pricing team to swiftly rectify the issue, earning **recognition with the "Premier Award" for cross-functional collaboration and rapid problem-solving.**
- Developed a semi-supervised machine learning model to identify fraudulent pharmacies, employing t-SNE for pharmacy clustering and XGBoost with Shapley values to establish cluster personas. The findings from this work were forwarded to the special investigative unit for necessary action.

Data Scientist - Machine Learning, Mercy Health

Feb 2018 - Sep 2021

- Built and deployed a machine learning model to predict DRG codes of inpatient billing based on clinical notes and discrete patient data. The production model **generates more than \$10 million in annual revenue** by identifying and correcting incorrectly coded cases early in the billing process. At the company-wide innovation conference, this project was **selected as a finalist (in the cost and efficiency category).**
- Automated the cancer registry case-finding process by using NLP techniques to imaging/pathology notes and physician notes. This resulted in savings by reducing the labor cost of finding cancer registry cases.
- Used predictive analytics to estimate the financial impact of elective surgery cancellations due to COVID-19. The findings were presented to C-suite leadership to aid with their decision-making process.

SKILLS

- **Programming Languages:** Python, SQL, Mathematica, MATLAB, C++, Java
- **Data Manipulation & Visualization:** Pandas, NumPy, Matplotlib, Seaborn, Tableau, QlikView, QlikSense
- **Machine Learning & Generative AI:** Scikit-Learn, XGBoost, LightGBM, Keras, Tensorflow, Pytorch, RAG, LLM Agents, LangChain

EDUCATION

Doctor of Philosophy, Physics

Washington University in St. Louis, Missouri, USA

Aug 2011 - Aug 2017

Bachelor of Science, Physics and Mathematics

Brigham Young University, UT, USA

Jan 2007 - Aug 2011

MACHINE LEARNING CERTIFICATIONS

- [Building Basic Generative Adversarial Networks](#)
- [Causal Inference from Observational Data](#)
- [Reinforcement Learning Specialization \(4 courses\)](#)
- [Introduction to TensorFlow for AI, ML, and DL](#)
- [Deep Learning Specialization \(5 Courses\)](#)